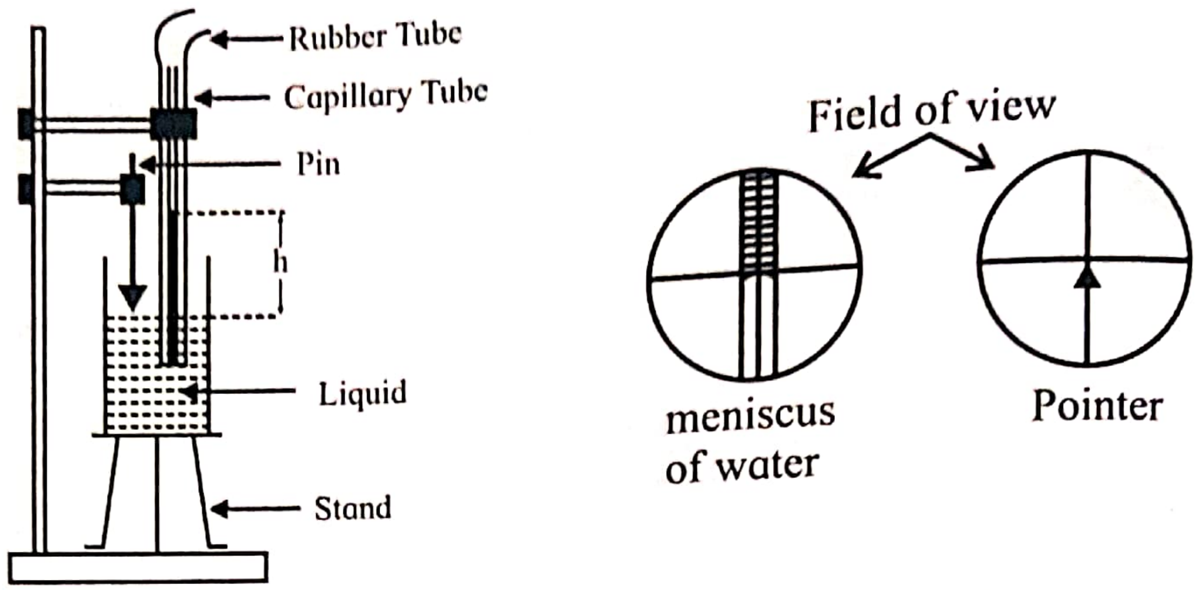
ACTIVITY NO. 1
EFFECT OF DETERGENT ON SURFACE TENSION

Aim : To study the effect of detergent on surface tension of water by capillary rise method.

Apparatus : Travelling microscope, beaker, index pin, stand, water, detergent powder.

Formula : $T = \dfrac{hr\rho g}{2\cos\theta}$

Diagram :



Procedure :

1. Determine the least count of travelling microscope.
2. Immerse the capillary tube vertically in the beaker containing water.
3. Remove the air bubbles by pressing the rubber tube. Adjust the level of water in the beaker so that meniscus in the bore of capillary should be above the edge of the beaker and below the lower edge of the capillary stand.
4. Adjust the tip of the pointer so that it just touches the level of water. Focus the travelling microscope on the meniscus such that the horizontal cross - wire is tangential to the convex surface of the meniscus. Note the reading on vertical scale (h_1) of travelling microscope
5. Remove the beaker without disturbing the position of the capillary tube.
6. Adjust the microscope such that the horizontal cross - wire is tangential to the tip of the pointer. Record the reading on the vertical scale (h_2) of travelling microscope
7. Repeat the steps 2 to 6 for solution of detergent powder .

Observations:

L. C. of travelling microscope:

Smallest division of main scale (P)	0.05	cm
Total no. of divisions on the Vernier scale (N)	50	
L.C. of travelling microscope (P/N)	0.001	cm

Observation:

Measurement of capillary rise (h):

Liquid	Reading when the microscope is focused on		$h = (h_1 - h_2)$ cm
	Meniscus h_1 cm	Tip of the pointer h_2 cm	
Water	2.5	0.5	h_w = 2
Detergent Solution	2	0.5	h_d = 1.5

Calculation : $h = (h_1 - h_2)$ cm

Result : 1) h_w = ...2........ cm

2) h_d = ...1.5........ cm

Conclusion : Rise of water in capillary is greater than rise of soap solution in capillary. Therefore Surface tension of water is greater than that of soap solution. Thus surface tension decreases due to the addition of detergent.

Precautions :

1. The capillary should be thoroughly cleaned.
2. There should be no air bubbles inside the capillary. Capillary must be immersed in the liquid.
3. The capillary should be vertical.

Questions

1. Why surface tension of a liquid decreases due to presence of detergent ?

Detergent acts as a surfactant and reduces the attractive forces between water molecules at the surface. As a result, the surface tension of the liquid decreases.

2. What is the role of detergent in washing clothes ?

Detergent lowers the surface tension of water and helps it penetrate the fabric more easily. It loosens and removes dirt, grease, and oil from clothes, making them clean.

Remark and sign of teacher : 26/9/22 A